REVIEW-REPORT

GOLDEN: Guideline-grounded Orchestrated LLM Dispatch for Emergency Networks

An India-Grounded, Guideline-Compliant Decision-Support Framework for Pre-Hospital Emergency Coordination Using Orchestrated LLM Agents

Orchestration Framework: LangGraph (Python)
Interoperability Standard: HL7 FHIR R4 / ABDM

Team Members

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Review 1 – 5 Marks

No.	Evaluation Parameter	Marks	Assessment Focus	Mapped CO(s)
1	Project title and technical relevance	1	Specific, meaningful and aligned with the selected domain/problem.	CO2, CO4
2	Abstract and problem definition	1	Clearly states the context, problem and need for the proposed work.	CO2, CO4
3	Project objectives and expected outcome	1	Objectives and expected outcomes are clearly defined and achievable.	CO2, CO4
4	Scope and feasibility	1	Achievable within the semester and available resources.	CO1, CO4
5	Preliminary work plan	1	Presents an initial timeline, milestones and allocation of responsibilities.	CO1, CO3, CO4
	Total	**5**		

1 Project Title and Technical Relevance

Project Title

GOLDEN: Guideline-grounded Orchestrated LLM Dispatch for Emergency Networks
(An India-Grounded, Guideline-Compliant Decision-Support Framework for Pre-Hospital Emergency Coordination Using Orchestrated LLM Agents)

Technical Relevance

The project sits at the intersection of **multi-agent AI orchestration** and **healthcare interoperability standards**, and is deliberately aligned with the Indian emergency-care domain rather than a generic international framing. Its technical relevance rests on the following points:

- **Multi-agent orchestration:** The system is built on LangGraph, a stateful, cyclic graph-based orchestration framework, using an orchestrator–specialist pattern with a shared, strongly-typed state object (Pydantic v2), conditional edges, retry cycles, and durable checkpointing – representing current, production-relevant practice in agentic AI system design rather than a simple prompt-chaining pipeline.

- **Standards-compliant interoperability:** The system is grounded in HL7 FHIR R4, the same standard underlying India's Ayushman Bharat Digital Mission (ABDM). Patient pre-registration is implemented as an actual FHIR bundle (`Patient` + `Encounter` + `Condition` resources) served via a HAPI FHIR server, making the project directly relevant to India's national health-data interoperability push rather than a closed, proprietary data model.

- **India-grounded problem framing:** Unlike comparable capstone projects that default to a US-centric frame (911, ESI triage, HIPAA), this project is explicitly re-anchored on India's 108/112 emergency response services, ABDM/ABHA, and the MoRTH Cashless Treatment of Road Accident Victims Scheme, 2025 – making the technical design choices verifiable against real, current Indian regulatory and clinical standards.

- **Applied benchmarking against an established research baseline:** The triage/EHR-interaction agent is benchmarked against MedAgentBench, a peer-reviewed multi-agent clinical-agent benchmark (Jiang et al., *NEJM AI*, 2025), situating the project's technical claims against a verifiable, published baseline rather than only self-reported metrics.

- **Deterministic safety engineering alongside LLM reasoning:** The design combines a deterministic, rule-based "hard-SOS" bypass with LLM-based reasoning – a technically relevant contribution because it directly addresses a known failure mode of LLM-based systems (unpredictable latency and hallucination) in a safety-critical setting.

2 Abstract and Problem Definition

Abstract

India records the world's highest absolute road-crash death toll, with the National Crime Records Bureau's *Accidental Deaths & Suicides in India (ADSI) 2024* report citing approximately 1.99 lakh traffic-related deaths in 2024 (up from 1.98 lakh in 2023) – averaging 546 deaths per day. Despite this scale, the "golden hour" following a crash is routinely lost to fragmented, manual coordination between India's 108 ambulance service, the 112 Emergency Response Support System (ERSS), and hospitals. This project proposes a **multi-agent AI decision-support system** that parallelizes emergency dispatch triage, hospital and bed selection, FHIR-based patient pre-registration, and autonomous family notification – tasks that are today performed serially and manually by a human dispatcher over the phone. The system is implemented as a stateful, cyclic LangGraph orchestration of a focused four-agent core (Coordinator, Triage/Dispatcher, Hospital & Bed, and Family-Notification Voice Agent), grounded in Indian clinical and regulatory standards (108/112, ABDM/FHIR R4, the May 2025 Golden-Hour Cashless Treatment Scheme), and is explicitly positioned as decision support for human dispatchers rather than autonomous medical decision-making.

Problem Definition

The economic burden of India's road-crash toll is severe: MoRTH estimates the socio-economic cost at approximately INR 1,47,114 crore (0.77% of GDP), rising to an estimated INR 5,96,820 crore (3.14% of GDP) once adjusted for underreporting, per the World Bank report *Traffic Crash Injuries and Disabilities: The Burden on Indian Society*. Peer-reviewed studies indicate that fewer than one in four road-traffic-accident victims reach a medical facility within the golden hour, and highway-based studies report emergency care access times of 60 minutes or greater in the majority of cases. This gap was formally recognized by the Supreme Court of India in *S. Rajaseekaran v. Union of India & Ors.* (2025 INSC 45, decided 8 January 2025), which held that the golden-hour provision under Section 162 of the Motor Vehicles Act upholds the right to life under Article 21, and directed the Central Government to frame an implementing scheme – leading to MoRTH's Cashless Treatment of Road Accident Victims Scheme, 2025, effective 5 May 2025.

Despite this policy push, India's emergency response remains fragmented across the 108 ambulance service, the 112 ERSS, and hospitals that are only beginning to interoperate through ABDM's FHIR R4 standard. In practice, the minutes between a crash and hospital admission are lost because a **human dispatcher** must triage the call, separately locate a hospital with an available and appropriate bed, separately notify the victim's family, and separately initiate patient registration paperwork – each step typically conducted over the phone, one after another, rather than in parallel. This project's core problem statement is therefore: *there is currently no lightweight, standards-compliant, India-grounded system that parallelizes emergency dispatch triage, hospital/bed selection, FHIR-based patient pre-registration, and family notification using coordinated AI agents, while remaining safe, auditable, and explicitly scoped as decision support rather than autonomous clinical decision-making.*

3 Project Objectives and Expected Outcome

3.1 Primary Objectives

P1. Design and implement a stateful, cyclic multi-agent orchestration (LangGraph) that ingests an emergency report, performs guideline-grounded acuity triage, recommends the optimal receiving hospital with bed availability, pre-registers the patient as a FHIR bundle, and triggers a family-notification voice call – end to end, within a defined target latency.

P2. Design and build an AI voice-calling agent for "Family Notification & Information-Gathering," bridging Exotel telephony with the Gemini Live conversational model through a custom Python audio bridge, integrated as an asynchronous node using webhook-based state resumption within the orchestration graph.

P3. Ground triage in Indian and standard clinical protocols, and demonstrate a deterministic hard-SOS bypass that guarantees life-threatening cases skip LLM inference latency.

3.2 Secondary Objectives

S1. Handle code-mixed (Tamil–English) emergency input and measure the accuracy delta relative to English-only input.

S2. Quantify and mitigate cascading hallucination across agent handoffs, using schema validation and inter-agent verification.

S3. Demonstrate model tiering / offline fallback (a hosted 70B model versus a locally served 7–8B model) under simulated degraded connectivity.

S4. Benchmark the triage/EHR-interaction agent against MedAgentBench.

3.3 Expected Outcomes

1. A working, end-to-end multi-agent decision-support system that demonstrably reduces the number of sequential manual steps between an emergency report and (a) a triaged, guideline-grounded acuity assessment, (b) a recommended hospital with confirmed simulated bed availability, (c) a written FHIR pre-registration bundle, and (d) an autonomous family notification call.

2. Quantitative evidence – via a set of ablation experiments (E1–E6) – that the multi-agent architecture outperforms a single-agent baseline on triage accuracy, hallucination rate, and/or latency, along with a measured accuracy delta for code-mixed Tamil–English input relative to English-only input.

3. A demonstrated deterministic hard-SOS bypass that reliably routes life-threatening cases around LLM inference latency, and a working degraded-connectivity fallback using a locally served small language model.

4. A benchmark score for the system's EHR-interaction agent on MedAgentBench, reported alongside the published baseline scores (Claude 3.5 Sonnet v2, GPT-4o, DeepSeek-V3, and open-weight models) for comparison.

5. A standards-compliant artifact – a FHIR R4-based patient pre-registration flow directly relevant to ABDM – serving as a strong viva talking point and a differentiator from US-centric (911/HIPAA) student projects.
6. A rehearsed, live demonstration script covering an Indian-context incident, parallel hospital/triage coordination, the live voice call, the hard-SOS bypass, and connectivity-loss resilience.
7. Explicit documentation of ethical and regulatory positioning: the system is framed throughout as decision support rather than autonomous medical decision-making, uses only synthetic patient data, and restricts all automated voice calls to consented team members under simulated scenarios.

4 Scope and Feasibility

4.1 In Scope

LangGraph-based orchestration; a Triage Agent; a Hospital/Bed Agent operating over a FHIR server; FHIR patient pre-registration; family voice-call integration; a live monitoring dashboard; an evaluation harness with ablation studies; India-grounded guideline prompting; a safety/guardrail validation layer; and a simulated ambulance/traffic visualization.

4.2 Out of Scope

Real patient data of any kind; real, unconsented call recipients; production hospital APIs; autonomous (non-human-reviewed) medical diagnosis; production ABDM certification (sandbox environment only); real traffic-signal control (SUMO simulation only, optional); and VLM-based medical trauma diagnosis (limited to accident detection only, optional).

4.3 Feasibility Within Semester and Available Resources

The project was originally scoped as a 7-agent system, which was assessed as over-scoped for a 3–4 person team within a single semester – demo value is concentrated in 3–4 agents, so the design has been deliberately reduced to a **focused four-agent core**, with two further agents (route/ambulance visualization and accident-detection vision) treated as optional stretch goals attempted only if the team is ahead of schedule. This reduction is the single largest feasibility correction applied to the plan.

A second major feasibility risk was the original intention to self-host 70B-class open-weight models (Llama-3.3-70B / Qwen-2.5-72B) via vLLM on student hardware. This is not achievable on typical student hardware – self-hosting such models requires roughly two 80GB GPUs (or four 48GB cards at 4-bit quantization), which a single consumer GPU (8–24GB) cannot provide, and this approach would fail at demo time. The plan has therefore been revised to use free-tier hosted inference (Groq's free tier for Llama 3.3 70B, Google Gemini 2.5 Flash's free tier, and OpenRouter free models) for the primary reasoning agents, with a small local model (Llama 3.1 8B or Qwen 2.5 7B) served via Ollama purely for the offline-fallback demonstration – which is feasible on a laptop-class GPU.

Remaining feasibility considerations:

- **Data access:** High-value clinical datasets (e.g., MIMIC-IV-ED) require PhysioNet credentialing and CITI training, which involves an external approval turnaround – this is initiated at the very start of the project to avoid blocking later phases.
- **Regulatory sandbox access:** ABDM sandbox registration involves an external approval turnaround and is initiated early in the Foundations phase.
- **Telephony/voice calling:** The AI voice-calling agent is planned as one of the project's strongest live-demo assets, built on Exotel's WebSocket-based bidirectional voice-streaming applet for outbound telephony, bridged to Google's Gemini Live conversational model via a custom Python service. This pairing is technically verified and compatible: Exotel streams raw PCM telephony audio at 8 kHz, while Gemini Live natively accepts 16-bit PCM input at 16 kHz and returns 16-bit PCM output at 24 kHz over its own WebSocket session – the Python bridge's resampling stages (8 kHz $\rightarrow$ 16 kHz on the way in, 24 kHz $\rightarrow$ 8 kHz on the way out, using `numpy`/`scipy`) map directly onto this format mismatch, and downstream call-result automation is handled via `n8n`

webhooks. Feasibility is bounded by TRAI's TCCCPR 2018 regulations: the demonstration is restricted to consented team phones running synthetic scenarios, since production-scale automated calling would require DLT telemarketer registration that is out of scope for an academic prototype – this constraint applies regardless of the specific telephony/voice-model pairing used. A further constraint is that Gemini Live's free-tier quota (concurrent sessions and per-session audio duration) is considerably tighter than the free tier used for the project's text-based reasoning agents, and is accounted for when scheduling live-demo rehearsals.

- **Optional agents as a safety valve:** Route/ambulance visualization and the accident-detection vision agent are explicitly kept optional, governed by a decision gate: if the core system is stable once the integration phase is complete, at most one optional agent is added; if not, all optional agents are cut in favor of deeper evaluation.

Taken together, the reduced agent count, the replacement of infeasible local model hosting with free-tier hosted inference, and the staged/optional treatment of higher-risk components make the project's core deliverables achievable within the semester using resources realistically available to a student team (free-tier API access, a consumer-grade GPU, and Dockerized open-source infrastructure).

5 Preliminary Work Plan

The project is planned for a 3–4 person team, structured into five phases, each with a clear focus, defined milestones, and a concrete deliverable marking the end of that phase.

Phase	Focus, Milestones & Deliverable
Phase 0 – Foundations	**Focus & Milestones:** Initiate PhysioNet credentialing and CITI training applications (all members); register for the ABDM sandbox; obtain Groq/Gemini/OpenRouter API keys; stand up HAPI FHIR in Docker and load synthetic Synthea patient data; freeze the problem statement, taxonomy, and architecture diagram. **Deliverable:** Working FHIR datastore, functioning LLM API calls, project document v1, and a presentation skeleton.
Phase 1 – Core Agents	**Focus & Milestones:** Build the Triage Agent with guideline prompts and Pydantic schemas; build the Hospital/Bed Agent with FHIR queries and pre-registration writes; build the LangGraph orchestrator with state management, checkpointing, and the hard-SOS bypass; design and build the voice-calling agent and integrate it via webhook-based asynchronous resumption. Responsibilities are allocated across team members by agent. **Deliverable:** An end-to-end "happy path" – text input through triage, hospital selection, FHIR write, and a fired voice call.
Phase 2 – Integration, Safety & Dashboard	**Focus & Milestones:** Add the guardrail/validation layer and inter-agent verification for hallucination mitigation; build the live dashboard (state, map, call status); implement the code-mixed input path with Whisper STT; add model-tiering / offline fallback via Ollama. **Deliverable:** Integrated demo v1 with an internal dry run.
Phase 3 – Evaluation & Experiments	**Focus & Milestones:** Build the evaluation harness; run ablation experiments E1–E6, including the MedAgentBench benchmark run; attempt optional agents (route visualization, SUMO, vision) only if the core system is already stable. **Deliverable:** Results tables and ablation charts.
Phase 4 – Writing & Polish	**Focus & Milestones:** Draft the research paper for arXiv submission; finalize the written report and presentation; rehearse the demo script with a contingency buffer for failures. **Deliverable:** Submission-ready paper and final viva package.

Decision Gate

Once the core four-agent system is stable and the Phase 1–2 deliverables are complete, the team adds exactly one optional agent (route visualization is recommended over SUMO or the vision agent). If the core is not yet stable at that point, all optional agents are cut and the remaining effort is reinvested into evaluation depth.

Change Triggers

- If free-tier LLM rate limits are hit during rehearsal, add a small OpenRouter credit top-up to raise the daily request floor and pre-cache demo runs in advance.
- If code-mixed (Tamil–English) accuracy proves too low, narrow the scope to Tamil–English only and report this as a finding rather than a failure.
- If MCP (Model Context Protocol) integration slips, it is dropped, since it is scoped as a stretch goal rather than a dependency.

6 Literature Review

Fifteen sources were reviewed across four thematic clusters: (i) multi-agent emergency dispatch and clinical decision-support architectures, (ii) FHIR/EHR-interaction benchmarks for LLM agents, (iii) India-specific emergency-medical-service (EMS) evidence, and (iv) LLM-based triage safety and multi-agent reliability. Together, these clusters motivate the project's architecture, its choice of evaluation metrics, and the specific research gap it targets.

6.1 Multi-Agent Emergency Dispatch and Clinical Decision Support

Li et al. (2026) present *DispatchMAS*, a taxonomy-grounded multi-agent emergency dispatch simulator built on AutoGen, in which LLM caller and dispatcher agents conduct structured calls across 32 chief-complaint categories and a six-phase protocol. Evaluated on 100 simulated dispatch cases derived from de-identified MIMIC-III records, it achieved 94% correct potential-agent contact and 97% correct call-back advice, with inter-rater agreement (Gwet AC1) above 0.70. This is the closest direct precedent for the project's Dispatcher/Triage Agent and its structured, protocol-constrained call flow; however, it is an English-oriented proof of concept with no real ambulance outcomes and no India-specific or code-mixed input, which this project addresses directly.

Ibrahim et al. (2026) present *TriAgent*, an adaptive multi-agent architecture for crisis clinical decision support under incomplete information, combining an orchestrator, retrieval sub-agent, clinical assessment and medication/allergy checks, a critique agent, and parallel safety verification. Evaluated on 1,000 MIMIC-IV-ED presentations with synthesized incomplete information, it reported 85.0% critical-case recall and 65.7% overall triage accuracy, substantially outperforming matched single-model/retrieval baselines (at most 14.7% critical-case recall). Its combination of adaptive orchestration, retrieval, critique, and an explicit safety layer is architecturally close to this project's own Coordinator + specialist-agent design, and is used as the primary architectural benchmark for comparison, with this project's contribution being an India-grounded emergency-coordination focus (triage *and* hospital/bed selection *and* family notification *and* registration) rather than clinical triage alone.

6.2 FHIR / EHR-Interaction Benchmarks

Jiang et al. (2025) introduce *MedAgentBench*, an interactive benchmark in which LLM agents must retrieve information and perform clinically meaningful actions inside a FHIR-compliant virtual EHR, rather than answering static questions. Across 300 clinician-written tasks and 100 virtual patient profiles, Claude 3.5 Sonnet v2 achieved the strongest reported overall success rate (69.67%), ahead of GPT-4o (64.00%) and Gemini 1.5 Pro (62.00%). This is the primary precedent for benchmarking this project's Hospital/FHIR Agent, and the project reuses MedAgentBench directly (Experiment E6) to situate its EHR-interaction agent against a published, reproducible baseline.

Lee et al. (2025/2026) present *FHIR-AgentBench*, which separates FHIR resource retrieval from answer generation across 2,931 clinician-sourced question-answer pairs derived from MIMIC-IV-FHIR. The strongest baseline reached only about 50% answer correctness, and multi-turn retrieval (71% recall) substantially outperformed single-turn retrieval (58% recall) – evidence that naive single-shot FHIR querying is inadequate and that full-context prompting over large FHIR bundles is impractical. This directly informs the Hospital/Bed Agent's design, which issues targeted, iterative FHIR queries rather than attempting to load entire patient records into context.

6.3 Clinical Multi-Agent Benchmarking Beyond Static QA

Schmidgall et al. (2026) present *AgentClinic*, a multimodal benchmark in which doctor, patient, measurement, and moderator agents interact sequentially under incomplete information, across nine specialties and seven languages. Reducing the number of allowed interactions from 20 to 10 dropped diagnostic accuracy from roughly 52% to 25%, and a persistent "Notebook" memory tool improved a Llama-3 agent's performance by up to 92% (relative). This paper is the strongest justification in the reviewed set for evaluating the project's agents *interactively* – including a Tamil–English multilingual experiment – rather than relying only on static, single-turn question answering.

6.4 India-Specific Emergency Medical Service Evidence

Gaikwad et al. (2025) retrospectively studied 315 road-traffic-accident victims at a tertiary hospital in Sangli, India, finding that only 20.6% reached care within the golden hour, with transport problems (33%), financial barriers (30.5%), unawareness of nearby facilities (14.6%), and language issues (13.3%) cited as the leading obstacles. This is the project's strongest India-specific empirical motivation, and its obstacle table maps directly onto the coordination failures the proposed system targets (transport/routing, hospital awareness, and language-aware interaction).

Jena et al. (2024) examined Maharashtra's 108 EMS system, where ambulance-based doctors also provide general consultations; the state recorded 935,544 such consultations in 2022, a 452% increase over 2020, demonstrating the scale of existing Indian EMS infrastructure that an AI coordination layer could augment. Modi et al. (2018) surveyed 1,220 respondents in Maharashtra and found that while 76.2% recognized the 108 number, only 20.2% had ever called it, and 82.9% preferred a unified emergency number – underscoring that any technical solution must also account for public awareness and usability. Together, these two studies support the project's explicit framing as a layer that *augments* existing 108/112 infrastructure and human dispatchers, rather than replacing them.

6.5 LLM-Based Triage: Feasibility and Safety Evidence

Shekhar et al. (2025) tested a general-purpose LLM (ChatGPT-4o Mini) against a critical-care paramedic panel on 392 real ambulance requests, finding 76.5% agreement overall and 93.8% agreement when the panel was unanimous – direct evidence that LLM-assisted prioritization is feasible, though the study evaluates relative ranking only, not end-to-end latency or patient outcomes.

Set against this, Cui et al. (2026) conducted a systematic review and meta-analysis of 11 studies (3,088 cases) on LLM emergency-department triage, finding pooled sensitivity of only 61% (95% CI 48–73%) for the highest-acuity category, despite high specificity (97%) – with very high heterogeneity (I^2=97%) across studies. This is the key paper motivating the project's **deterministic hard-SOS bypass**: it demonstrates that LLM-only triage cannot be trusted to reliably catch the most critical cases, so critical-case detection must not depend on LLM inference alone.

Shen et al. (2025) show that knowledge-grounded, ESI-instruction-tuned models (their MIETIC/SDTA approach, best demonstrated on Qwen2.5-72B) achieve near-perfect high-risk recall (1.00) and strong overall accuracy (0.91), supporting the value of guideline grounding over open prompting – directly motivating Experiment E2 (guideline-grounded vs. open prompting). Wang et al. (2025) and Gaber et al. (2025) provide further precedent for evaluating triage, referral, and RAG-assisted clinical workflows using GPT-4o/GPT-4-Turbo and Claude 3.x respectively, reinforcing that prompt design and retrieval grounding are experimental variables worth measuring explicitly rather than fixed choices.

Finally, Walonoski et al. (2018) introduce *Synthea*, the open-source synthetic patient simulator (with native FHIR/C-CDA export) that underlies this project's synthetic patient datastore, allowing safe demonstration and testing without exposing real patient information.

6.6 Multi-Agent Reliability and Cascading Errors

Lin et al. (2026) introduce *AgentAsk*, which identifies four recurring failure modes at agent-to-agent handoffs – Data Gap, Signal Corruption, Referential Drift, and Capability Gap – and proposes a lightweight clarification mechanism that improved accuracy by up to 4.69% while keeping added latency and cost under 10% of baseline overhead. This is the main paper underlying the project's cascading-hallucination research gap (Experiment E1) and directly motivates the Pydantic schema-validation and inter-agent verification layer used at every hand-off between the Coordinator and its specialist agents.

6.7 Summary Table

No.	Title (short)	Authors, Year	Key Result	Relevance to Project
1	DispatchMAS	Li et al., 2026	94% correct agent contact; 97% correct callback advice; Gwet AC1 >0.70	Precedent for Dispatcher/Triage Agent and structured call flow
2	MedAgentBench	Jiang et al., 2025	Claude 3.5 Sonnet v2: 69.67% overall success on FHIR-based EHR tasks	Primary benchmark for the Hospital/FHIR Agent (Experiment E6)
3	AgentClinic	Schmidgall et al., 2026	Accuracy drops from ∼52% to ∼25% with fewer interaction turns	Justifies interactive, multi-turn, multilingual evaluation
4	FHIR-AgentBench	Lee et al., 2025/26	Best baseline ∼50% answer correctness; multi-turn beats single-turn retrieval	Informs iterative, targeted FHIR querying design
5	Golden Hour in RTA Victims	Gaikwad et al., 2025	Only 20.6% reached care within the golden hour; transport/finance/language barriers	Core India-specific problem-statement evidence
6	Maharashtra 108 EMS Access	Jena et al., 2024	935,544 consultations in 2022; 452% growth over 2020	Confirms scale of existing EMS to augment, not replace

No.	Title (short)	Authors, Year	Key Result	Relevance to Project
7	Public Awareness of EMS	Modi et al., 2018	76.2% knew 108; only 20.2% had ever called it	Motivates usability/awareness-aware design
8	LLM Ambulance Dispatch & Triage	Shekhar et al., 2025	76.5% agreement with paramedic panel (93.8% when unanimous)	Feasibility precedent for the Triage Agent
9	LLM ED Triage Meta-Analysis	Cui et al., 2026	Pooled sensitivity only 61% for highest-acuity triage	Motivates the deterministic hard-SOS bypass
10	Knowledge-Embedded LLM Triage	Shen et al., 2025	Qwen2.5-72B: 1.00 high-risk recall with guideline grounding	Motivates guideline-grounded prompting (Experiment E2)
11	LLM ED Triage & Guidance	Wang et al., 2025	GPT-4-Turbo: 100% MEWS accuracy after prompt engineering	Precedent for triage and hospital-routing evaluation
12	LLM Clinical Decision Workflows	Gaber et al., 2025	Claude 3.5 (clinical): 64.40% exact triage; RAG workflow 65.75%	Supports RAG-grounded, multi-decision workflow design
13	Synthea	Walonoski et al., 2018	Large-scale synthetic patient generation with FHIR export	Underlies the project's synthetic FHIR datastore
14	AgentAsk	Lin et al., 2026	Up to 4.69% accuracy gain from handoff clarification, $<10\%$ overhead	Basis for schema validation / inter-agent verification (E1)
15	TriAgent	Ibrahim et al., 2026	85.0% critical-case recall vs. $\leq 14.7\%$ for baselines	Closest architectural benchmark for the four-agent design

6.8 Research Gap

Across all fifteen sources, no single study combines: (a) an India-grounded emergency-response context with documented, India-specific access barriers (transport, cost, awareness, and language); (b) native handling of code-mixed Tamil–English emergency speech; (c) a multi-agent architecture that *paral-*

lelizes triage, hospital/bed selection, FHIR-based pre-registration, and family notification, rather than addressing only clinical triage (DispatchMAS, TriAgent) or only FHIR/EHR retrieval (MedAgentBench, FHIR-AgentBench) in isolation; (d) a deterministic, non-LLM-dependent bypass for the highest-acuity cases, directly motivated by the safety gap Cui et al. (2026) expose in LLM-only triage; and (e) explicit, measured mitigation of cascading errors across agent handoffs using the AgentAsk (Lin et al., 2026) failure taxonomy. This project is positioned to fill that combined gap, and its planned ablation experiments (E1–E6, defined in the Proposed Methodology) are each traceable to a specific paper reviewed above.

7 System Architecture

7.1 Architectural Pattern

The system follows an **orchestrator–specialist multi-agent pattern**, implemented as a stateful, cyclic graph in LangGraph. A single Coordinator agent owns a shared, strongly-typed state object (Pydantic v2) and routes execution to specialist agents via conditional edges. The graph supports retry cycles, `interrupt()`-based human-in-the-loop breakpoints, and durable checkpointing (SQLite/PostgreSQL), which is what allows the graph to pause for minutes while an asynchronous voice call is in progress and then resume exactly where it left off.

7.2 Component Overview

Coordinator / Orchestrator

Receives the incoming emergency report, performs the deterministic hard-SOS check first (bypassing all LLM calls for unambiguous life-threatening cases), then fans work out to the Triage Agent and Hospital/Bed Agent in parallel, merges their outputs, and triggers the Voice Agent.

Triage / Dispatcher Agent

Parses the incident report (text or Whisper-transcribed speech, including Tamil–English code-mixed input), assigns an acuity level using guideline-grounded prompting, and returns a schema-validated triage object.

Hospital & Bed Agent

Queries the FHIR server (HAPI FHIR, pre-loaded with Synthea synthetic data) for candidate hospitals, checks a simulated bed-availability table, ranks candidates by acuity and distance, and writes the FHIR pre-registration bundle (`Patient` + `Encounter` + `Condition`).

Voice Agent (Family Notification & Info-Gathering)

An Exotel outbound call is triggered via webhook; a Python audio bridge resamples audio between Exotel's 8 kHz telephony stream and the Gemini Live model's 16 kHz input / 24 kHz output, while `n8n` handles posting the final call outcome back to the orchestrator, resuming the paused graph.

Safety / Validation Layer

Wraps every agent's output in Pydantic schema gates and deterministic rule checks, plus a single hosted moderation call, directly motivated by the AgentAsk handoff-failure taxonomy discussed in the Literature Review.

Dashboard

A live view of in-flight cases, agent state, and call status, used both for demonstration and as a human-in-the-loop review point.

7.3 Data Flow

1. An incident report (text or transcribed speech) enters the Coordinator.
2. The Coordinator performs the hard-SOS check. Life-threatening cases are routed immediately to the Hospital & Bed Agent, skipping LLM-based triage latency entirely.

3. For all other cases, the Triage Agent and Hospital & Bed Agent run in parallel – triage against the guideline knowledge base, and hospital/bed lookup against the FHIR server – rather than serially, as a human dispatcher would.

4. The Coordinator merges both outputs, writes the FHIR pre-registration bundle, and asynchronously triggers the Voice Agent.

5. The graph checkpoints and pauses; execution resumes when the Voice Agent's call-completed webhook returns (allergies, medications, consent, blood group, etc.).

6. The Safety/Validation Layer checks every hand-off along the way; the Dashboard reflects state changes in real time.

7.4 Architecture Diagram

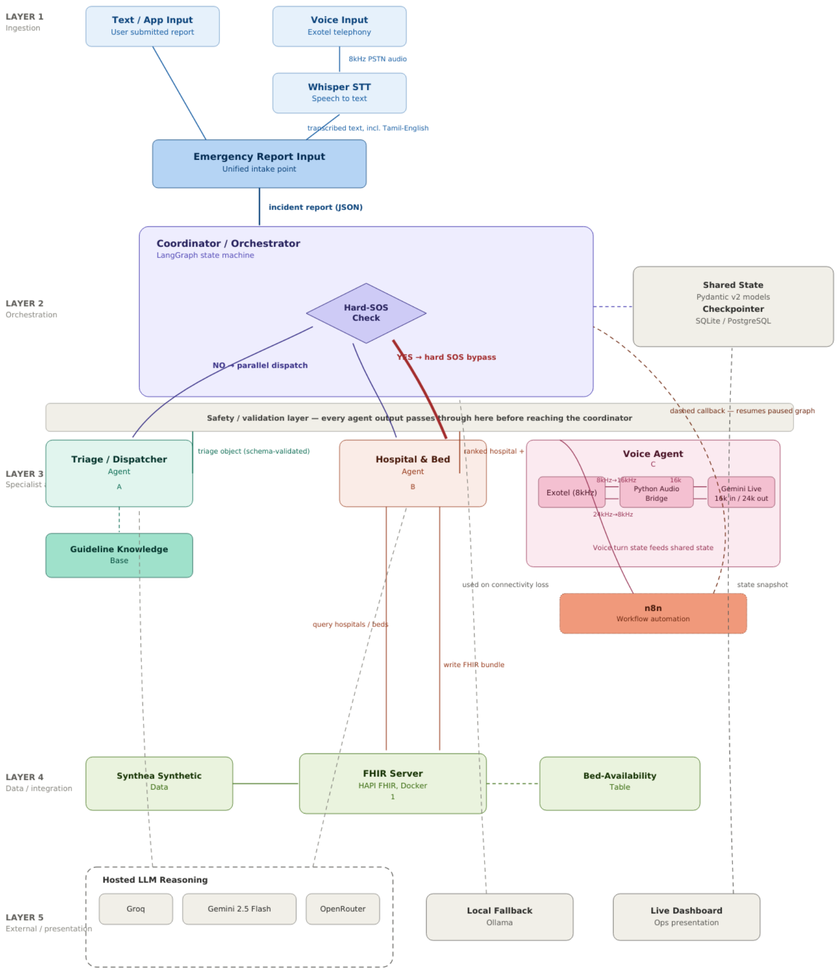

Figure 1: System architecture

8 Design of Proposed Methodology and System Design

8.1 Design Philosophy and Guiding Principles

The design of GOLDEN is governed by four principles, each chosen to directly address a gap or risk identified in the Literature Review:

- **Decision support, not replacement.** The system augments a human dispatcher's parallel-processing capacity; it never issues a final clinical or dispatch decision without a human review point (via LangGraph's `interrupt()` breakpoints), consistent with the project's ethical positioning stated throughout this report.
- **Safety-by-design over probabilistic-only reasoning.** Because Cui et al. (2026) show LLM-only triage misses up to 39% of the most critical cases, the highest-acuity path is handled by a deterministic rule engine *before* any LLM is invoked, not as a post-hoc check on an LLM's output.
- **Fail-safe degradation over hard failure.** Every external dependency (hosted LLM APIs, the voice/telephony stack) has a defined fallback path (local model tiering, logged failed callbacks) so a single component outage degrades the system's capability rather than stopping it entirely.
- **Auditability by construction.** Every inter-agent hand-off is schema-validated and logged, directly motivated by the AgentAsk handoff-failure taxonomy (Data Gap, Signal Corruption, Referential Drift, Capability Gap) discussed in the Literature Review, so failures are traceable to a specific agent and a specific failure mode rather than surfacing as an opaque end-to-end error.

8.2 Proposed Methodology

The project's methodology combines **deterministic rule-based engineering** for safety-critical decisions with **LLM-based agentic reasoning** for open-ended interpretation (free-text/speech triage, guideline application, conversational information-gathering), coordinated through a stateful multi-agent graph. This hybrid approach is deliberately evaluated, not assumed: every major design choice is paired with a controlled ablation experiment, so that the panel and any subsequent reviewer can see *evidence* for why each component is included, rather than taking the architecture on faith.

Ablation and Evaluation Experiments

E1 – Core Architecture Ablation

Single-agent (one LLM performs the entire pipeline end to end) versus the full multi-agent pipeline, compared on triage accuracy/F1, hallucination rate, and latency. Tests whether multi-agent decomposition is actually justified, motivated by DispatchMAS and TriAgent's reported gains over single-model baselines.

E2 – Prompt Grounding Ablation

Guideline-grounded prompting versus open/unconstrained prompting for the Triage Agent, compared on triage F1 and high-risk recall. Directly tests the finding from Shen et al. (2025), where guideline grounding drove high-risk recall from well below 1.00 to 1.00.

E3 – Safety Bypass Ablation

The deterministic hard-SOS path versus an LLM-only decision path for the highest-acuity cases, compared on latency and critical-case miss rate. Directly tests the safety gap identified by Cui et al. (2026) (61% pooled sensitivity for highest-acuity LLM triage).

E4 – Multilingual Robustness
English-only input versus Tamil–English code-mixed input, measuring the resulting triage accuracy delta and the Whisper word-error-rate (WER) increase. Addresses the language barrier identified as a contributing factor in 13.3% of golden-hour failures by Gaikwad et al. (2025).

E5 – Tiering / Offline Resilience
A hosted 70B-class reasoning model versus the locally served 7–8B fallback model, compared on output quality, latency, and availability under simulated connectivity loss.

E6 – External Benchmark Validation
The Hospital & Bed (EHR-interaction) Agent evaluated on MedAgentBench, with results reported directly against its published baseline scores (Claude 3.5 Sonnet v2, GPT-4o, DeepSeek-V3), so the project's own benchmark claim is independently comparable rather than self-defined.

8.3 System Design

Shared State Design

All agents read from and write to a single, strongly-typed Pydantic v2 state object owned by the Coordinator – this is the backbone that lets the graph pause for an asynchronous voice call and resume correctly. Its design groups fields into five categories:

- **Case identity & input:** case ID, raw input (text or transcript), detected language, input modality.
- **Triage output:** acuity level, hard-SOS flag, confidence score, guideline reference used.
- **Hospital/FHIR output:** ranked hospital candidates, selected hospital, bed-confirmation status, FHIR bundle ID.
- **Voice/family-notification output:** call status, consent flag, collected information (allergies, medications, blood group), retry count.
- **Control & audit:** current graph node, validation errors from the Safety Layer, checkpoint ID, timestamps for each stage transition (used later for latency evaluation in E1–E3).

Per-Agent Algorithmic Design

Hard-SOS Rule Engine
A deterministic, non-LLM keyword and pattern matcher checked first against a fixed, reviewed list of unambiguous critical indicators (e.g., unresponsive, not breathing, no pulse, severe uncontrolled bleeding, cardiac arrest). Design goal: zero LLM calls, sub-second latency, and a false-negative rate close to zero even at the cost of some false positives (which merely route a case through the fast path unnecessarily rather than causing harm).

Triage Agent
Constructs a guideline-grounded prompt (drawing on a curated Indian/standard clinical guideline knowledge base), generates a structured output constrained to a fixed JSON schema (acuity

level, confidence, rationale), and rejects/retries any output that fails schema validation before it is written to shared state.

Hospital & Bed Agent
Issues targeted, iterative FHIR queries (informed by the FHIR-AgentBench finding that multi-turn retrieval outperforms single-shot retrieval) rather than loading full patient bundles into context; ranks candidate hospitals with a weighted scoring function over acuity match, distance, and live bed availability; writes the FHIR pre-registration bundle as an idempotent, retry-safe transaction so a network retry cannot create duplicate patient records.

Voice Agent
Fires the outbound call as a non-blocking, asynchronous task; defines a fixed webhook contract for the call-completion callback (status, transcript, extracted fields); on callback failure, persists the payload for manual recovery rather than silently dropping it, and resumes the checkpointed graph exactly at the node where it paused.

Safety / Validation Layer
Applied uniformly at every hand-off: (1) Pydantic schema gate, (2) deterministic rule checks (e.g., value ranges, required fields), (3) a single hosted moderation call. A failure at any stage routes back to the originating agent for a bounded number of retries before escalating to a human-in-the-loop breakpoint, rather than propagating a bad value forward.

Design Patterns Employed

- **Orchestrator–worker pattern:** the Coordinator owns control flow and shared state; specialist agents are stateless with respect to anything not explicitly passed to them, which keeps each agent independently testable.
- **Circuit-breaker / fallback pattern:** hosted-model calls that fail or exceed a latency threshold trigger a fallback to the locally served model, rather than blocking the whole pipeline.
- **Idempotent-writer pattern:** FHIR bundle writes and voice call triggers are designed to be safely retryable without creating duplicate patients or duplicate outbound calls.
- **Checkpoint-and-resume pattern:** the LangGraph checkpointer persists state at every node transition, which is what allows the graph to survive a multi-minute pause during the voice call and resume deterministically from the webhook callback.

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